

Indian Institute of Technology Indore
Science and Technology Council
IITISoC 2026

Project Proposal

Domain: DYNAMO
Dynamics, Modelling & Optimisation of Physical Systems

Track	Track 2 — Computational Simulation
PS Name	PS 2 — Biomechanical Fingerprinting: From Tissue Modelling to Synthetic Gel Design
	(Preference 1)
Difficulty	Advanced

Submitted by Team

Abhishek Nigam (Team Leader)

May 2026

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1. Team Details

1.1 Team Leader

Name Abhishek Nigam
Roll Number 240008001
GitHub github.com/2x-coder
LinkedIn linkedin.com/in/abhisheknigam8001
Skills Python, LAMMPS, OVITO, $\text{\LaTeX}$, MATLAB

1.2 Other Team Members

Field	Details
Member 2 — Samarth Dhanuka	
Roll Number	240008011
GitHub	github.com/TheRealSam-21
LinkedIn	linkedin.com/in/dhanuka-samarth
Skills	Python, Scientific Computing, SciPy, NumPy, LAMMPS
Member 3 — Kavyansh Raj Singh	
Roll Number	240008014
GitHub	github.com/kavyansh29
LinkedIn	linkedin.com/in/kavyansh-raj-singh
Skills	Python, Molecular Dynamics, Computational Simulation, matplotlib
Member 4 — Jayshil Mendhe	
Roll Number	240008013
GitHub	github.com/jayship-619
LinkedIn	linkedin.com/in/jayshil-mendhe
Skills	Python, Finite Element Analysis, Data Visualisation, Git

2. Problem Statement Details

Domain	DYNAMO
Track	2 — Computational Simulation
PS Name	Biomechanical Fingerprinting: From Tissue Modelling to Synthetic Gel Design
Difficulty	Advanced

3. Project Solution

3.1 Overview

In silico" refers to experiments or research conducted entirely through computer simulations and computational models.

Biological tissues such as the **myocardium and arterial walls** are highly non-linear, hyperelastic, and anisotropic materials. Their mechanical behaviour is characterised by a unique stress–stretch relationship that encodes information about the tissue’s microstructure. By extracting this *material fingerprint*, it becomes possible to reverse-engineer synthetic **polymer hydrogels** *in silico* whose mechanical response faithfully mimics that of the biological tissue — a critical capability for the development of surgical simulators, soft robotics, and tissue engineering scaffolds.

Our approach integrates three well-established computational tools into a single end-to-end pipeline: *as also provided in the problem statement*

1. **Experimental data mining** from peer-reviewed biomechanics literature.
2. **Automated material fingerprinting** using the open-source Python pipeline by Flaschel, Kumar & De Lorenzis (CMAME 2021) to fit hyperelastic constitutive models to experimental stress–stretch data.
3. **In-silico gel design and tensile testing** in LAMMPS to match the fingerprinted tissue response.

We will focus on **cardiac tissue (myocardium)** as our primary target, as it combines fibre-reinforced anisotropy with pronounced non-linearity and is extensively studied in the literature — making both data acquisition and model validation straightforward.

3.2 Task 1 — Literature Mining and Data Extraction

We will systematically search biomechanics journals (e.g. *Journal of the Mechanical Behavior of Biomedical Materials*, *Acta Biomaterialia*, *Biomechanics and Modeling in Mechanobiology*) for experimental uniaxial and biaxial stress–stretch (σ – λ) datasets for **human** or **porcine myocardium**. Data will be digitised using WebPlotDigitizer where raw numerical data is unavailable, and stored in a consistent CSV format for downstream processing. *Heart muscle of pigs*

3.3 Task 2 — Material Fingerprinting

We will deploy the Flaschel fingerprinting pipeline to fit the extracted stress–stretch data to candidate hyperelastic constitutive models. The pipeline tests multiple model families including:

- **Mooney–Rivlin** — two-parameter phenomenological model suitable as a baseline.

- **Demiray** — exponential model capturing the pronounced stiffening seen in soft biological tissues.
- **Holzapfel–Gasser–Ogden (HGO)** — fibre-reinforced model with two fibre families, best suited for anisotropic cardiac tissue.

The model with the lowest residual error across loading modes will be selected. The resulting material constants (e.g. c_1 , c_2 for Mooney–Rivlin; μ , k_1 , k_2 , κ for HGO) constitute the **material fingerprint** of the tissue.

3.4 Task 3 — In-Silico Gel Construction in LAMMPS

A three-dimensional coarse-grained crosslinked polymer network will be assembled in LAMMPS to represent a synthetic hydrogel:

- Polymer chains will be modelled using a bead-spring (FENE) framework on a tetrahedral or random-walk backbone.
- Crosslinks will be introduced at prescribed densities to tune the network’s elastic modulus.
- The initial configuration will be generated using LAMMPS’s built-in chain-building tools and then equilibrated under the NPT ensemble.

3.5 Task 4 — Biomechanical Mimicry via Iterative Optimisation

Simulated tensile tests will be performed using `fix deform` in LAMMPS. The virial stress tensor will be computed at each deformation step to obtain the simulated σ – λ curve. The architectural parameters — specifically **crosslink density** and **chain length between crosslinks** — will be systematically varied in a grid search to minimise the L_2 distance between the simulated gel curve and the fingerprinted tissue curve:

$$\mathcal{L} = \int_{\lambda_{\min}}^{\lambda_{\max}} [\sigma_{\text{gel}}(\lambda) - \sigma_{\text{tissue}}(\lambda)]^2 d\lambda$$

The final gel architecture that best replicates the tissue fingerprint will be reported along with overlay comparison plots.

3.6 Key Deliverables

- Curated and digitised experimental σ – λ dataset for myocardial tissue.
- Output logs and validation plots from the Flaschel fingerprinting pipeline, with material constants tabulated.
- LAMMPS data files and input scripts for gel construction, equilibration, and tensile testing.
- Comparative overlay plot of biological tissue data vs. LAMMPS gel response.
- Full GitHub repository with documented code and a technical report in $\text{\LaTeX}$.

4. Project Schedule

Period	Dates	Milestones
Week 1–2	3–16 Jun 2026	Literature review; identify and digitise 3–5 experimental σ – λ datasets for myocardial tissue. Set up Flaschel pipeline locally. Repository initialised.
Week 3–4	17–30 Jun 2026	Run fingerprinting pipeline on all datasets. Select best-fit constitutive model; extract and validate material constants. Prepare mid-term progress report. (<i>Mid-Term Evaluation: 30 Jun 2026</i>)
Week 5–6	1–14 Jul 2026	Build and equilibrate LAMMPS coarse-grained polymer network for three initial parameter sets. Implement <code>fix deform</code> tensile-testing script; validate virial stress computation.
Week 7–8	15 Jul–1 Aug 2026	Grid search over crosslink density and chain length; iteratively optimise gel architecture to minimise $\mathcal{L}$. Generate overlay comparison plots. (<i>IITISoC Hackathon window: 25 Jul–1 Aug 2026</i>)
Week 9–10	3–17 Aug 2026	Final documentation; write technical report in L ^A T _E X; record demo. Final submission and presentation. (<i>Final Submission Window: 3–17 Aug 2026</i>)

5. Tech Stack

- **LAMMPS** — Molecular dynamics engine for gel construction and tensile testing.
- **Python** (NumPy, SciPy, matplotlib) — Data processing, fingerprinting pipeline, and visualisation.
- **Flaschel et al. Pipeline** — Automated hyperelastic model fitting (<https://github.com/Material-Fingerprinting/material-fingerprinting-hyperelasticity>).
- **OVITO** (optional) — Visualisation of polymer network configurations.
- **Git / GitHub** — Version control and collaboration.
- **L^AT_EX** — Report writing.